

Technological Humanities

Jennifer L. Lieberman • Anto Mohsin
Editors

Technological Humanities

Essays in Honor of Ronald R. Kline

palgrave
macmillan

Editors

Jennifer L. Lieberman
Community College of
Allegheny County
Pittsburgh, PA, USA

Anto Mohsin 
Northwestern University in Qatar
Doha, Qatar

ISBN 978-3-032-24740-7

ISBN 978-3-032-24741-4 (eBook)

<https://doi.org/10.1007/978-3-032-24741-4>

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use. The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Palgrave Macmillan imprint is published by the registered company Springer Nature Switzerland AG.

The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

*This book is dedicated to Ronald R. Kline, scholar, co-author, teacher,
and mentor.*

ACKNOWLEDGEMENTS

This volume grew out of conversations (some long-running and others serendipitous) that reflect the kinds of intellectual communities Ronald R. Kline has spent a lifetime cultivating. It is therefore fitting that *Technological Humanities* takes shape not simply as a tribute to Ron's scholarship but as an expression of the relationships, mentorships, and shared commitments that his work has inspired. We dedicate this book to Ron with deep gratitude and admiration, in recognition of the profound impact he has had as a scholar, teacher, editor, and mentor. His insistence that technologies be understood through the lives, values, and agency of people, especially those often overlooked in dominant narratives, has shaped entire fields and continues to guide how many of us think, teach, and write.

We are especially thankful to Ron for his generosity in agreeing to be interviewed for this volume and for taking the time to reflect with us on his intellectual journey, his teaching, and his continuing engagement with questions about technology and society. That conversation—thoughtful, candid, and grounded in storytelling—captures the spirit of this book and models the kind of humanistic attention to technology that Ron has long championed. It was a privilege for us both, as editors and former students of his work, to learn from him once again in dialogue.

The idea for this collection began, fittingly, in conversation. On February 16, 2024, Anto spoke with Ray and Suzanne in Evanston, Illinois. Over coffee, he shared a long-held desire to edit a collection in honor of Ron, one that would not only celebrate his scholarship but also reflect the range of work his mentorship has made possible. Ray

immediately suggested reaching out to Jenni Lieberman, noting that she, too, had been thinking along similar lines. Jenni readily accepted the invitation to collaborate, and from that moment the project became a shared endeavor in the fullest sense. The ease with which this collaboration formed speaks directly to the intellectual generosity and collegial culture Ron himself has modeled.

We extend our sincere thanks to the contributors to this volume who represent Ron's co-authors, advisees, and even "grand student": Clinton J. Andrews, Lina Moe, Jamey Wetmore, Christina Dunbar-Hester, Ray Fouché, Nathan Kapoor, Suzanne Moon, Ling-Fei Lin, Heather A. Love, and Honghong Tinn. Each of these scholars brings a distinctive voice, methodology, and set of concerns to the volume, yet all share a commitment to understanding technology as a human practice that is historically situated, politically contested, and socially meaningful. Their chapters engage Ron's work in diverse ways: by extending his insights into new empirical domains, by revisiting familiar themes through fresh analytical lenses, and by carrying forward his emphasis on users, teachers, and communities as central actors in technological change. We are grateful not only for the excellence of their contributions but also for their patience, responsiveness, and generosity throughout the editorial process.

We also thank the editorial team at Palgrave Macmillan. Rachel Ballard and Karthika Devi Ravikumar who guided this project with care and enthusiasm from its earliest proposal through the final stages of manuscript preparation. Their support made it possible for us to shape the book thoughtfully and deliberately, and their confidence in the project sustained it through the inevitable complexities of editing an interdisciplinary collection.

This project was nurtured within supportive institutional environments, and we gratefully acknowledge the Community College of Allegheny County and Northwestern University in Qatar for providing the conditions that allowed this work to unfold. Though different in context and mission, both institutions value teaching, critical inquiry, and engaged scholarship, and they have shaped how we think about the role of the humanities in technological societies.

Finally, we wish to acknowledge the broader networks of colleagues, students, and interlocutors whose informal and unrecorded conversations have influenced this book. Ron's legacy reminds us that scholarship is not only produced in monographs and articles but also in classrooms, advising meetings, conference hallways, and shared moments of intellectual

curiosity. If this volume succeeds in foregrounding those human dimensions of technological inquiry, it is because it stands within a tradition of mentorship and community that Ron helped build and that continues to flourish through the work of those gathered here.

CONTENTS

1	Critical Humanism Confronts Technology	1
	Jennifer L. Lieberman and Anto Mohsin	
2	Generative Discursive “Disunity”: Nobert Wiener’s Disciplinary Agility as Contemporary Rhetorical Toolkit	19
	Heather A. Love	
3	The Strange Story of Automotive Collabs and Oil Embargoes	49
	Rayvon Fouché	
4	Accommodating Urban Micromobility Innovations	71
	Clinton J. Andrews and Lina Moe	
5	Consumers at the Coastline: The Camping Tent as Deviant Technology in Twenty-First-Century Los Angeles	101
	Christina Dunbar-Hester	
6	Developing Dispossession: Retrofitting Settler Colonialism at the Huntly Power Station	129
	Nathan Kapoor and Suzanne Moon	

7	Manufacturing Plus: Sociotechnical Imaginaries, Interpretative Flexibility, and the Ascent of Taiwan's Semiconductor Foundry Model	155
	Ling-Fei Lin	
8	Teaching the History of Emerging Technologies in an Information Age	187
	Honghong Tinn	
9	Science Outside the Lab: Ethics Training Through Science Policy Immersion	219
	Jameson M. Wetmore	
10	Teaching and Learning Technology Through Stories: A Conversation with Ronald R. Kline	239
	Anto Mohsin and Jennifer L. Lieberman	

NOTES ON CONTRIBUTORS

Clinton J. Andrews directs the Center for Urban Policy Research at Rutgers University, where he is Distinguished Professor of Urban Planning and Policy Development. He was educated at Brown University and Massachusetts Institute of Technology in Engineering and Planning, and he worked previously in the private sector and at Princeton University. He performs research on how people use the built environment, addressing how technological and environmental changes affect urban life. His books include *Humble Analysis: The Practice of Joint Fact-Finding*, *Regulating Regional Power Systems* and *Industrial Ecology and Global Change*. He is a Fellow of AAAS stands for American Association for the Advancement of Science and IEEE is an abbreviation for Institute of Electrical and Electronics Engineers.

Christina Dunbar-Hester is the author of *Oil Beach: How Toxic Infrastructure Threatens Life in the Ports of Los Angeles and Beyond* (2023); *Hacking Diversity: The Politics of Inclusion in Open Technology Cultures* (2020); and *Low Power to the People: Pirates, Protest, and Politics in FM Radio Activism* (2014); as well as numerous popular and academic articles and book chapters. Dunbar-Hester received her doctoral degree in Science and Technology Studies (STS) from Cornell University. She is Professor of Communication in the University of Southern California's Annenberg School.

Rayvon Fouché holds a joint appointment as Professor of Communication Studies in the School of Communication and Professor of Journalism in the Medill School of Journalism, Media, and Integrative Marketing Communications at Northwestern University, where he also directs the Humanity and Technoscience (HAT) Lab. He received his Ph.D. in Science and Technology Studies (STS) from Cornell University. He is the author of *Black Inventors in the Age of Segregation* (2003), *Game Changer: The Technoscientific Revolution in Sports* (2017), and *Technoskepticism: Between Possibility and Refusal* (2025).

Nathan Kapoor is Assistant Professor of History at Illinois State University, where he teaches classes in the History of Science, Technology, and Medicine. His forthcoming book, *Powering Colonialism*, explores the history of the relationship between settler-colonialism and electrification in New Zealand.

Jennifer L. Lieberman is Associate Dean of Humanities, Social Science, and Mathematics at the Community College of Allegheny County in Pittsburgh, Pennsylvania. After completing her doctorate in English at the University of Illinois, Urbana-Champaign, she was awarded a National Science Foundation postdoctoral fellowship to study Science and Technology Studies (STS) at Cornell University. She is the author of *Power Lines: Electricity in American Life and Letters, 1882–1952* (MIT Press, 2017), as well as over a dozen essays on the nexus of literary studies and STS, appearing in venues ranging from *Feminist Encounters through Configurations* and *Transatlantic Currents: Essays in Honor of David E. Nye*.

Ling-Fei Lin received her doctoral degree in Science and Technology Studies (STS) from Cornell University in 2015, specializing in the history of Taiwan's PC design–manufacturing nexus and high-tech production. She conducted postdoctoral research at Nanyang Technological University (2016–2017) in Singapore and was a postdoc researcher and Asian Center Fellow at Harvard University (2018–2019). Lin's research interests have grown to encompass digital Egyptology since 2019. After studying ancient Egyptian languages and cultures, she is currently pursuing a dual Ph.D. in Informatics and Egyptology at Indiana University Bloomington, where she also teaches as a Ph.D. instructor or associate instructor.

Heather A. Love is Associate Professor of English at the University of Waterloo (Ontario, Canada), where she conducts interdisciplinary research

and teaches courses on topics related to literature, culture, technology, health, and engineering education. She is the author of *Cybernetic Aesthetics: Modernist Networks of Information and Data* (Cambridge University Press, 2023), which was awarded the Gertrude J. Robinson Book Prize from the Canadian Communication Association and was a finalist for the British Society for Literature and Science Book Prize. Her work has appeared in journals such as *Modernism/modernity*, *New Literary History*, *Health Humanities*, and *IEEE Technology and Society Magazine*.

Lina Moe is a Ph.D. candidate in Planning and Public Policy at Rutgers University. Her research examines how institutions govern emerging technologies, from social robots to large-scale solar infrastructure. She is also a Research Fellow at the Federal Reserve Bank of St. Louis.

Anto Mohsin is Associate Professor in Residence in the Liberal Arts Program at Northwestern University in Qatar. He is an STS scholar specializing in infrastructure, energy, disaster, and environmental studies, with a focus on Indonesia in the twentieth century and a broader interest in Southeast Asia. He authored *Electrifying Indonesia: Technology and Social Justice in National Development* (University of Wisconsin Press, 2023). He earned his doctorate in STS from Cornell University and was trained in the university's Southeast Asia Program.

Suzanne Moon is Luke N. Walker Presidential Professor in the Department of History of Science, Technology, and Medicine at the University of Oklahoma. She specializes in the history of technology in Southeast Asia, especially Indonesia. Her current work is focused on agriculture, climate change, and sociotechnical imaginaries that inform adaptation. She was awarded the 2021 Da Vinci prize from the Society for the History of Technology (SHOT).

Honghong Tinn is an associate professor of the Program in the History of Science, Technology, and Medicine and the Department of Electrical and Computer Engineering at the University of Minnesota Twin Cities. She received her Ph.D. in STS from Cornell University. Her publications have appeared in *Technology and Culture*, *Osiris*, *IEEE Annals of the History of Computing*, and *East Asian Science, Technology and Society*. She is the author of *Island Tinkerers: Innovation and Transformation in the Making of Taiwan's Computing Industry* (MIT Press, 2025) and a co-author of *Computer: A History of the Information Machine*, 4th ed. (Routledge, 2023).

Jameson M. Wetmore is an associate professor and Director of the Center for Engagement and Training in Science & Society at Arizona State University. He completed his dissertation on the socio-political history of automobile safety in the US under the direction of Ron Kline in 2003 at Cornell University. He continues to explore the ways in which society shapes technology and vice versa and has run 17 versions of Science Outside the Lab since 2007. He is an Associate Director for Societal and Ethical Implications of the National Nanotechnology Coordinated Infrastructure and a co-author of *Technology & Society: Building our Sociotechnical Future* (MIT Press, 2009).

LIST OF FIGURES

Fig. 2.1	Visual representation of the “elaborate lacework” design of current US rail infrastructure in the Midwest (“How U.S. Cities Can Prepare,” 81; image credit: Anthony Sodaro and Frank Stockman, <i>LIFE Magazine</i>).	30
Fig. 2.2	Cover image for the <i>Life Magazine</i> article, accompanied by the caption “Even in peacetime New York’s Fifth Avenue is paralyzed by congestion” (“How U.S. Cities Can Prepare,” 77; image credit: Andreas Feininger, <i>LIFE Magazine</i>).	41
Fig. 2.3	Line drawing of “the Wiener civil defense plan’s” proposal for a series of protective “outer belts” that will preserve movement of people and supplies in the wake of a nuclear attack (“How U.S. Cities Can Prepare,” 79; image credit: Anthony Sodaro, <i>LIFE Magazine</i>).	42
Fig. 4.1	Number of U.S. cities hosting shared micromobility services from 2015 to 2022 (Data Source: BTS 2024)	79
Fig. 4.2	Market-led adoption versus planning-led adoption	82
Fig. 5.1	The advertised half-dome tent (screenshot by author, March 19, 2025)	105
Fig. 5.2	A half-dozen polyester tents with small “Free Palestine” signs affixed to them, University of Southern California campus, April 26, 2024 (photo by Cristina Visperas)	107
Fig. 5.3	Camouflage four-person camping dome tent (screenshot by author, March 17, 2025)	109
Fig. 5.4	A ten-pack of tents for sale on Amazon, listing homelessness as one use (Screenshot by author, March 23, 2025)	112

- Fig. 7.1 Morris Chang's "Presentation on Developing Very Large Scale Integration Industry (發展超大型積體電路工業簡報" on September 10, 1985 ("74" is the year the Republic of China, which was established in 1911). Document provided by Ding-Hua Hu (photo by author, March 9, 2011) 175
- Fig. 7.2 A Comparative Table in Morris Chang's "Presentation on Developing Very Large Scale Integration" on September 10, 1985. It compares four IC companies, UMC, Vitelic, Quasel, Mosel, in terms of their organization types, source of technology, and products and technology strategies. Document provided by Ding-Hua Hu (photo by author, March 9, 2011) 176